

# NexusFinance — API & Database Reference

Digital Lending Platform for Cambodia

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|-----------------|-------------------------|
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[API Reference](#) • [Endpoints](#) • [Database Schema](#) • [Developer Guide](#)

# Table of Contents

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1. API Overview
2. Authentication Endpoints
3. Loan Endpoints
4. Transaction Endpoints
5. Admin Endpoints
6. KHQR Endpoints
7. PayWay Endpoints
8. Database Schema Detail

# 1. API Overview

Base URL: <https://nexusfinance-lof3.onrender.com/api>

All endpoints return JSON. Protected endpoints require a Bearer token in the Authorization header.

| Status Code | Meaning                                 |
|-------------|-----------------------------------------|
| 200         | Success                                 |
| 400         | Bad request (missing/invalid fields)    |
| 401         | Unauthorized (invalid or missing token) |
| 403         | Forbidden (insufficient role)           |
| 404         | Resource not found                      |
| 429         | Rate limited                            |
| 500         | Internal server error                   |

## 2. Authentication Endpoints

### POST /api/auth/register

Register a new user account.

| Field    | Type   | Required | Description         |
|----------|--------|----------|---------------------|
| name     | string | Yes      | User's full name    |
| email    | string | Yes      | Valid email address |
| password | string | Yes      | Min 6 characters    |
| phone    | string | No       | Phone number        |

Response: 201 { message: 'Account created. Please verify your email.' }

### POST /api/auth/login

Authenticate and receive a JWT token.

| Field | Type | Required | Description |
|-------|------|----------|-------------|
|-------|------|----------|-------------|

|          |        |     |                  |
|----------|--------|-----|------------------|
| email    | string | Yes | Registered email |
| password | string | Yes | Account password |

Response: 200 { token: 'jwt...', user: { id, name, email, role } }

Response (unverified email): 403 { error: '...', code: 'EMAIL\_NOT\_VERIFIED' } — verify your email via send-otp + verify-otp first.

## POST /api/auth/send-otp

Send a 6-digit OTP to the user's email.

| Field | Type   | Required | Description          |
|-------|--------|----------|----------------------|
| email | string | Yes      | Email to send OTP to |

## POST /api/auth/verify-otp

Verify the OTP code.

| Field | Type   | Required | Description   |
|-------|--------|----------|---------------|
| email | string | Yes      | Email address |
| code  | string | Yes      | 6-digit code  |

## 3. Loan Endpoints

### GET /api/loans

List all loans. Customers see only their own; admins see all.

Response: 200 [ { id, user\_id, name, email, amount, type, status, ... } ]

### POST /api/loans

Create a new loan application. Requires authentication.

| Field | Type   | Required |
|-------|--------|----------|
| name  | string | Yes      |

|                |        |     |
|----------------|--------|-----|
| email          | string | Yes |
| amount         | number | Yes |
| type           | string | Yes |
| purpose        | string | No  |
| monthlyIncome  | string | No  |
| durationMonths | string | No  |

## PATCH /api/loans/:id

Update a loan status. Requires loan-officer or super-admin role.

| Field  | Type   | Required                            |
|--------|--------|-------------------------------------|
| status | string | Yes<br>(approved/rejected/<br>hold) |

Dedicated actions: PATCH /api/loans/:id/approve, /api/loans/:id/reject, /api/loans/:id/hold — all restricted to loan-officer and super-admin.

## 4. Transaction Endpoints

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### GET /api/transactions

List transactions. Customers see their own; admins see all.

### POST /api/transactions

Record a new transaction.

### POST /api/simulate-payment

Simulate a payment callback (for testing). Triggers Telegram notification.

## 5. Admin Endpoints

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The following endpoints require a loan-officer or super-admin role:

| Endpoint        | Method    | Description                |
|-----------------|-----------|----------------------------|
| /api/stats      | GET       | Platform statistics        |
| /api/users      | GET       | List all users (paginated) |
| /api/audit/logs | GET       | Audit log entries          |
| /api/config     | GET/PATCH | Platform configuration     |
| /api/tasks      | GET       | KYC/review tasks           |

## 6. KHQR Endpoints

| Endpoint           | Method | Description                 |
|--------------------|--------|-----------------------------|
| /api/khqr/generate | POST   | Generate a KHQR code        |
| /api/khqr/verify   | POST   | Verify a payment by MD5     |
| /api/khqr/decode   | POST   | Decode a KHQR string        |
| /api/khqr/deeplink | POST   | Generate a payment deeplink |

## 7. PayWay Endpoints

| Endpoint                     | Method | Description                                              |
|------------------------------|--------|----------------------------------------------------------|
| /api/payway/generate-qr      | POST   | Create a PayWay payment and persist a PENDING record     |
| /api/payway/verify-payment   | POST   | Check transaction status; marks APPROVED + writes ledger |
| /api/payway/callback         | POST   | ABA webhook (HMAC-signed) with approval/decline updates  |
| /api/payway/simulate-payment | POST   | Sandbox-only helper to approve a payment for testing     |
| /api/payway/transactions     | GET    | List recent PayWay transactions from the database        |

Approved PayWay payments are written to nexus\_payway\_transactions and create a Repayment entry in nexus\_transactions, plus an in-app and Telegram notification.

## 8. Database Schema Detail

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### nexus\_users

| Column           | Type        | Description                                                   |
|------------------|-------------|---------------------------------------------------------------|
| id               | SERIAL PK   | Auto-increment ID                                             |
| name             | TEXT        | User's full name                                              |
| email            | TEXT UNIQUE | Email address                                                 |
| password         | TEXT        | Bcrypt password hash                                          |
| role             | TEXT        | customer / loan-officer / super-admin                         |
| phone            | TEXT        | Phone number                                                  |
| telegram_chat_id | BIGINT      | Telegram chat ID (if linked)                                  |
| email_verified   | BOOLEAN     | Whether the email has been verified (blocks login when false) |
| otp_code         | TEXT        | Bcrypt hash of OTP                                            |
| otp_expires_at   | TIMESTAMPTZ | OTP expiry time                                               |
| otp_verified_at  | TIMESTAMPTZ | When OTP was verified                                         |
| created_at       | TIMESTAMPTZ | Account creation time                                         |

### nexus\_loans

| Column  | Type      | Description                    |
|---------|-----------|--------------------------------|
| id      | SERIAL PK | Auto-increment ID              |
| user_id | INTEGER   | Applicant user ID              |
| name    | TEXT      | Applicant name (denormalized)  |
| email   | TEXT      | Applicant email (denormalized) |
| amount  | NUMERIC   | Loan amount                    |

|                 |             |                                      |
|-----------------|-------------|--------------------------------------|
| type            | TEXT        | Loan type (SME, Personal, etc.)      |
| status          | TEXT        | pending / approved / rejected / hold |
| purpose         | TEXT        | Loan purpose                         |
| duration_months | INTEGER     | Repayment duration                   |
| interest_rate   | NUMERIC     | Annual interest rate                 |
| monthly_income  | NUMERIC     | Applicant monthly income             |
| assigned_to     | INTEGER     | Assigned officer user ID             |
| created_at      | TIMESTAMPTZ | Application date                     |

## nexus\_payway\_transactions

| Column     | Type         | Description                   |
|------------|--------------|-------------------------------|
| id         | BIGSERIAL PK | Auto-increment ID             |
| tran_id    | TEXT UNIQUE  | PayWay transaction ID         |
| email      | TEXT         | Customer email (normalized)   |
| amount     | NUMERIC      | Payment amount                |
| currency   | TEXT         | Currency (default USD)        |
| status     | TEXT         | PENDING / APPROVED / DECLINED |
| apv        | TEXT         | PayWay approval code          |
| loan_id    | TEXT         | Linked loan ID (optional)     |
| user_id    | INT          | NexusFinance user ID          |
| created_at | TIMESTAMPTZ  | Creation time                 |
| paid_at    | TIMESTAMPTZ  | When the payment was approved |